

A Survey of Memory and Long-Context Mechanisms for Long-Horizon LLM Agents

InteractiveSurvey

Abstract

The rapid advancement of Large Language Models has enabled autonomous agents to execute complex, multi-step tasks, yet their long-horizon efficacy remains constrained by limited context windows and the inability to maintain coherence over unbounded interaction histories. Traditional static approaches often succumb to context drift and computational inefficiency, necessitating sophisticated mechanisms that emulate human-like memory retention and recall. This survey provides a comprehensive examination of memory and long-context architectures specifically designed for long-horizon agents, distinguishing between structural, lifecycle, and operational dimensions of agentic memory. We synthesize recent advancements to delineate the pivotal shift from passive context storage to active memory governance. Our analysis reveals that hierarchical and graph-based topologies significantly enhance retrieval precision by encoding causal relationships, while biologically inspired consolidation strategies effectively balance information fidelity with computational efficiency through adaptive forgetting. Furthermore, integrating agent-native operations, such as self-reflection and build-and-critique loops, transforms raw conversational data into curated knowledge units, thereby mitigating hallucinations and ensuring robust long-term reasoning. By offering a unified taxonomy and evaluating trade-offs regarding scalability and consistency, this work establishes a foundational reference for developing next-generation agents capable of sustaining reliable performance across indefinite interaction horizons.

1 Introduction

The rapid advancement of Large Language Models (LLMs) has catalyzed the development of autonomous agents capable of executing complex,

multi-step tasks over extended periods [1]. However, the efficacy of these long-horizon agents is fundamentally constrained by the limitations of standard context windows, which struggle to maintain coherence, factual consistency, and semantic relevance as interaction histories grow unbounded. Traditional approaches relying on static sliding windows or naive concatenation often succumb to context drift, the "lost in the middle" phenomenon, and prohibitive computational costs. Consequently, the ability to effectively manage, retrieve, and synthesize information across vast temporal spans has emerged as a critical bottleneck, necessitating sophisticated mechanisms that transcend simple token buffering to emulate human-like memory retention and recall.

This survey paper provides a comprehensive examination of memory and long-context mechanisms specifically designed for long-horizon LLM agents [2]. Unlike previous reviews that focus broadly on context extension techniques or general retrieval-augmented generation, this work zeroes in on the architectural and algorithmic innovations that enable agents to persist, reason, and adapt over prolonged interaction cycles. We investigate how modern systems transform unstructured conversational streams into structured, dynamic knowledge bases that support complex planning, causal inference, and persona consistency. By synthesizing recent advancements in hierarchical structuring, biologically inspired lifecycle management, and agent-native operations, this paper aims to delineate the current state-of-the-art and identify the pivotal shifts from passive context storage to active memory governance.

The first major segment of this review analyzes architectural designs for agentic memory, focusing on how structural complexity enhances retrieval precision and reasoning depth [3]. We explore hierarchical and graph-based topologies, including temporal tree architectures that segment

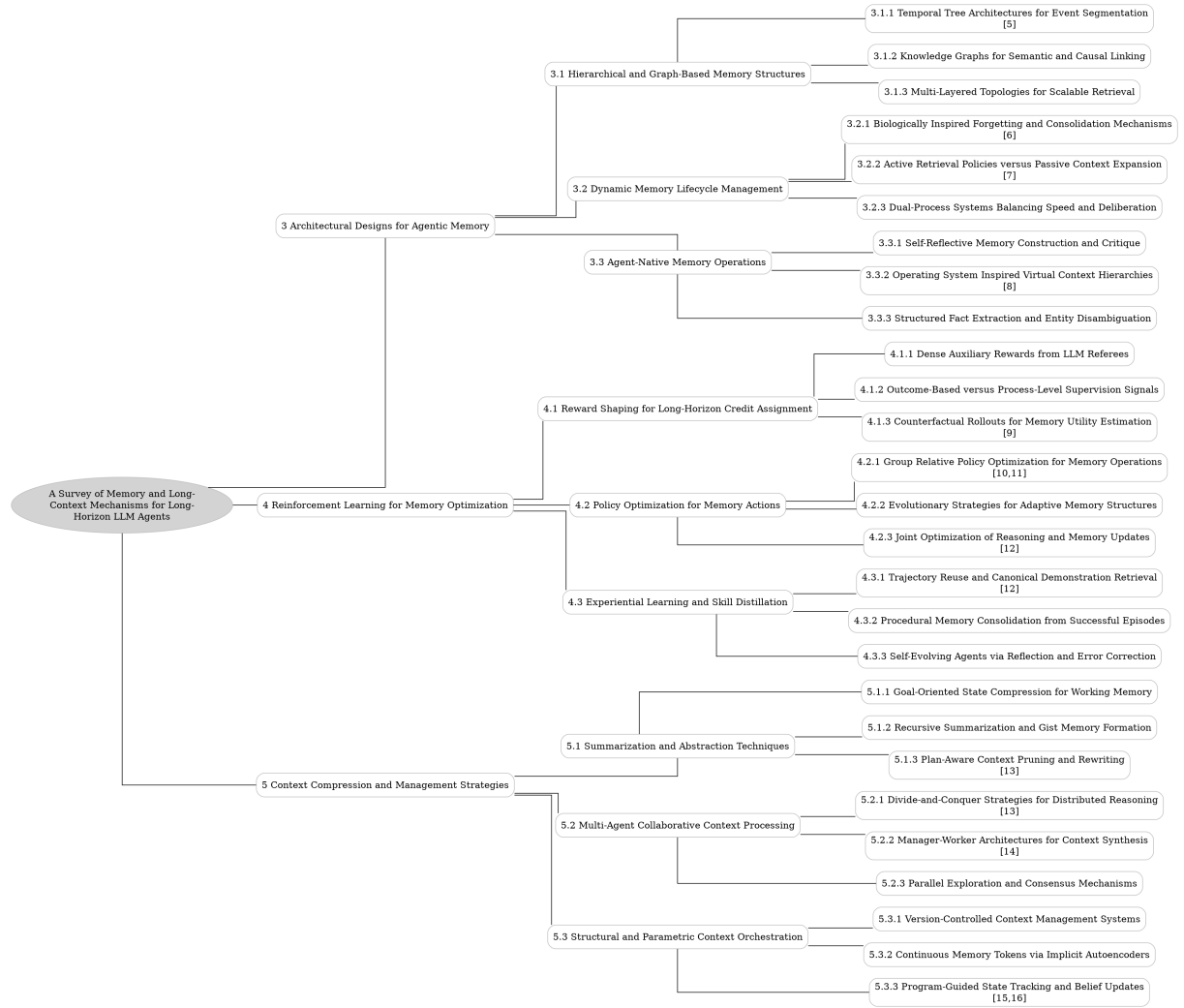


Figure 1: The outline of our survey: A Survey of Memory and Long-Context Mechanisms for Long-Horizon LLM Agents

$$I(t) = I_0 \cdot e^{-\lambda t}$$

Figure 2: Chart from 'Human-Inspired Memory Architecture for LLM Agents' (arXiv:2605.08538)

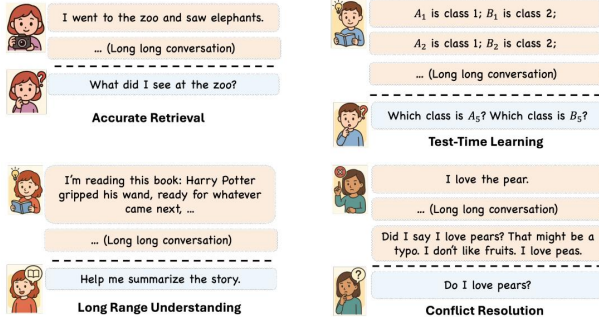


Figure 3: Chart from 'Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions' (arXiv:2507.05257)

events based on semantic boundaries rather than rigid time windows, and knowledge graphs that encode causal and semantic relationships to facilitate multi-hop inference. Furthermore, we examine multi-layered topologies that stratify memory into working, summarized, and archival tiers, enabling sub-linear retrieval complexity while balancing information fidelity with computational efficiency. These structural innovations collectively address the scalability challenges inherent in flat vector stores, ensuring that agents can navigate branching discourse trajectories without succumbing to semantic decay.

$$P_{ij} = \frac{\tilde{w}_{ij}}{\sum_k \tilde{w}_{ik}}$$

Figure 4: Chart from 'GAAMA: Graph Augmented Associative Memory for Agents' (arXiv:2603.27910)

Subsequently, the discussion shifts to dynamic memory lifecycle management, contrasting passive context expansion with active, biologically inspired consolidation strategies. We detail mechanisms that mimic hippocampal-neocortical interactions, where rapid episodic encoding transitions into stable semantic storage through adaptive for-

getting and recursive blending. This section critically evaluates the divergence between static, pre-processed embeddings and active retrieval policies that treat memory access as a controllable, iterative action. Additionally, we analyze dual-process systems that explicitly decouple rapid heuristic retrieval from slow, deliberative reasoning, optimizing the trade-off between latency and fidelity by dynamically allocating computational resources based on task complexity and memory sufficiency.

Finally, we investigate agent-native memory operations that integrate self-reflection and critique directly into the memory construction process. This involves examining "build-and-critique" loops where candidate memories are generated as rich narratives and subsequently validated for factual consistency and logical coherence before consolidation. By embedding meta-cognitive capabilities within the encoding phase, these systems transform raw conversational turns into curated knowledge units, effectively pruning noise and reinforcing high-fidelity episodic details. This evolution from passive logging to active curation ensures that memory repositories remain robust against hallucinations and redundancies, thereby supporting reliable long-term reasoning.

The primary contribution of this survey lies in its unified framework for categorizing and evaluating memory mechanisms tailored specifically for long-horizon autonomy. We provide a systematic taxonomy that distinguishes between structural, lifecycle, and operational dimensions of agentic memory, offering clear insights into the trade-offs of each approach regarding scalability, consistency, and computational overhead [4]. By synthesizing disparate lines of research into a cohesive narrative, this paper not only clarifies the current technological landscape but also highlights critical gaps in existing methodologies. Ultimately, this work serves as a foundational reference for researchers and practitioners aiming to develop next-generation agents capable of sustaining coherent, reliable, and efficient performance across indefinite interaction horizons.

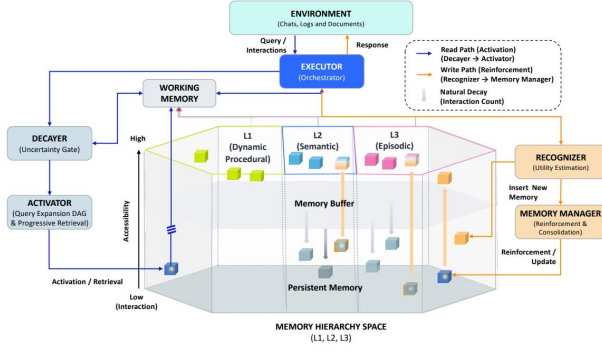


Figure 5: Chart from 'OBLIVION: Self-Adaptive Agentic Memory Control through Decay-Driven Activation' (arXiv:2604.00131)

2 Architectural Designs for Agentic Memory

2.1 Hierarchical and Graph-Based Memory Structures

2.1.1 Temporal Tree Architectures for Event Segmentation

Temporal tree architectures address the limitations of flat memory stores by organizing conversational history into hierarchical, chronologically grounded structures. Unlike linear sliding windows that risk context drift, these systems partition unbounded input streams into discrete episodic contexts using semantic boundary detection. This segmentation preserves the intrinsic temporal order of events, allowing the model to distinguish between short-term interaction details and long-term abstract patterns while maintaining constant memory complexity regardless of total conversation length.

The core mechanism involves constructing dynamic trees where nodes represent specific event states and edges encode causal or associative temporal relationships. By employing dual-channel segmentation, these architectures refine event granularity beyond simple time-stamping, integrating explicit structural representations with implicit semantic modeling. This approach enables the system to capture branching discourse trajectories, where a single event may belong to multiple traces, thereby reflecting the intersecting nature of real-world dialogue more accurately than rigid linear chains [5].

Consequently, temporal trees significantly enhance reasoning capabilities regarding entity state evolution and fact validity periods across sessions. The hierarchical design facilitates efficient

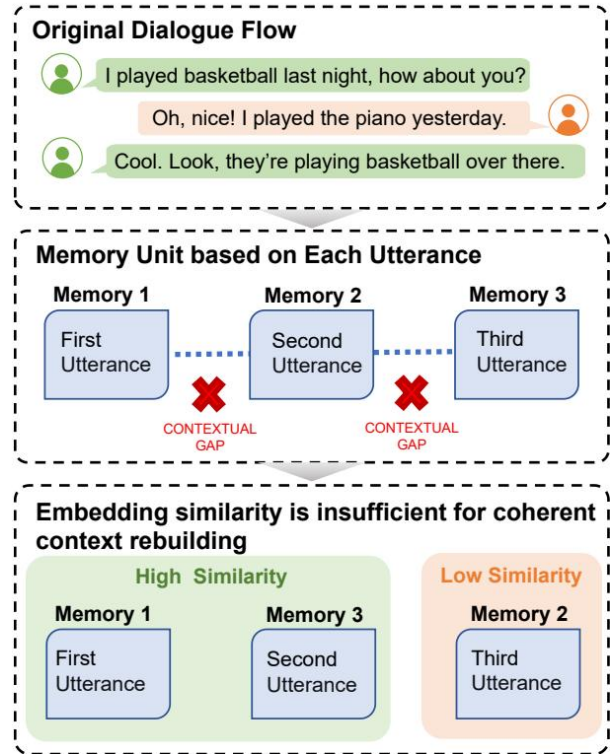


Figure 6: Chart from 'Membox: Weaving Topic Continuity into Long-Range Memory for LLM Agents' (arXiv:2601.03785)

retrieval of temporally adjacent yet semantically diverse events, crucial for answering complex queries about specific timeframes or causal dependencies. By decoupling planning from evidence-aware writing and utilizing global in-document retrieval over fine-grained units, these architectures mitigate noise and redundancy, ensuring that composition-critical evidence remains accessible for multi-hop reasoning tasks without succumbing to semantic decay.

2.1.2 Knowledge Graphs for Semantic and Causal Linking

Knowledge graphs serve as a foundational architecture for transforming unstructured conversational history into structured, retrievable memory units. Unlike flat vector embeddings that often suffer from destructive de-contextualization, graph-based approaches distill long-form interactions into atomic facts and semantic triplets. This process involves mapping source and destination entities into a latent space, where new nodes are created or merged with existing ones based on semantic similarity thresholds. By enforcing rigid structural schemas, these systems preserve the integrity of complex sequential dependencies, en-

abling efficient lookup while maintaining the nuanced relationships between disparate pieces of information that are critical for advanced reasoning.

Beyond static semantic linking, recent advancements integrate causal structures to support multi-hop inference and counterfactual planning. These frameworks construct subgraphs that explicitly model cause-and-effect relationships, allowing agents to navigate complex relational paths rather than relying solely on lexical overlap. Techniques such as damped spreading activation propagate queries along semantic, temporal, and causal edges, effectively retrieving records that are semantically distant but causally relevant. This capability is particularly vital for tasks requiring root cause analysis or debugging, where understanding the generative chain of events is more important than simple factual recall.

To ensure memory consistency over time, these architectures employ dynamic update mechanisms governed by temporal precedence and conflict resolution protocols. When new experiences trigger updates, the system evaluates existing nodes for redundancy using metrics like mutual information and Jaccard similarity before merging conflicting facts. Temporal metadata acts as a disambiguation layer, prioritizing recent updates while preserving historical context necessary for tracing reasoning chains. This synthesis of semantic anchoring, causal discovery, and temporal governance creates a robust memory substrate that supports faithful, deep reasoning across extended interaction horizons without succumbing to the limitations of fixed context windows.

2.1.3 Multi-Layered Topologies for Scalable Retrieval

Multi-layered topologies address the scalability limitations of flat vector stores by organizing memory into hierarchical structures that separate working, summarized, and archival layers. This architectural stratification enables sub-linear retrieval complexity by routing queries through coarse-grained semantic clusters before accessing fine-grained episodic details. Systems like Raptor and HiMem exemplify this approach, utilizing recursive abstraction to compress sequential contexts into tree-organized nodes or distinct note-and-episode tiers, thereby balancing information fidelity with computational efficiency during long-term interaction.

Advanced retrieval mechanisms within these

topologies employ adaptive gating and graph-guided traversal to bridge decoupled storage layers. Instead of relying solely on static similarity matching, agents dynamically adjust search depth and candidate set sizes based on query uncertainty and temporal relevance. By integrating multi-factor signals such as confidence scores, role-based attributes, and temporal continuity, these systems facilitate multi-hop reasoning across layers, ensuring that high-level thematic cues effectively ground the retrieval of precise atomic facts without exhaustive scanning of the entire memory space.

Despite improved latency and coherence, multi-layered designs introduce challenges in maintaining consistency across abstraction boundaries and managing the overhead of consolidation processes. The efficacy of such architectures is highly task-dependent; while they excel in persona consistency and complex historical recall, they may incur unnecessary computational costs for simple, localized queries where flat retrieval suffices. Future iterations must optimize the trade-off between hierarchical indexing maintenance and real-time inference speed, particularly as agents evolve to handle procedural knowledge alongside declarative facts in autonomous environments.

2.2 Dynamic Memory Lifecycle Management

2.2.1 Biologically Inspired Forgetting and Consolidation Mechanisms

Biologically inspired memory systems increasingly adopt the complementary learning systems theory, mimicking the hippocampal-neocortical interaction where rapid episodic encoding transitions into stable semantic storage. Unlike static retrieval architectures, these models implement dynamic consolidation processes that distinguish between transient conversational details and persistent user states. By separating recording time from real-world event occurrence, modern approaches generate durative summaries that capture long-term traits while preserving the temporal integrity essential for coherent reasoning across extended interaction horizons.

To prevent memory explosion and mitigate interference, these frameworks integrate adaptive forgetting mechanisms modeled after human decay curves and retrieval-induced forgetting [6]. Rather than aggressive compression that severs contextual dependencies, systems employ proba-

bilistic decay where unused traces naturally diminish while frequently accessed memories are refreshed or boosted. Recursive blending strategies utilize retention parameters to balance abstraction with detail, ensuring that redundant information is pruned without losing rare but vital facts, thereby maintaining a scalable memory footprint that evolves continuously with agent experience.

$$\text{Layer}(m_i) = \begin{cases} \text{LML} & \text{if } I_i(t) \geq \theta_{\text{promote}} \\ \text{SML} & \text{if } I_i(t) < \theta_{\text{demote}} \end{cases}$$

Figure 7: Chart from 'FADEMEM: BIOLOGICALLY-INSPIRED FORGETTING FOR EFFICIENT AGENT MEMORY' (arXiv:2601.18642)

Advanced implementations further refine this through gated recurrent structures and thermodynamic diffusion, treating memory as a dynamic field rather than a fixed database. These mechanisms enable selective updating where new observations trigger reconsolidation, allowing the system to correct contradictions and integrate novel skills without catastrophic forgetting. By aligning memory evolution with cognitive principles of chunking and associative binding, agents achieve improved retention persistence and reasoning stability, effectively bridging the gap between biological memory dynamics and artificial long-term context management.

2.2.2 Active Retrieval Policies versus Passive Context Expansion

Passive context expansion strategies, including massive window architectures and static compression, treat memory as a fixed substrate where reasoning is deferred entirely to inference time. While effective for preserving sequential continuity, these methods suffer from the "lost in the middle" phenomenon and incur prohibitive computational latency as sequence lengths scale. Furthermore, standard retrieval-augmented generation often inherits this passivity by relying on static, pre-processed embeddings that sever temporal dependencies, reducing rich episodic histories to isolated geometric points incapable of supporting complex multi-hop reasoning or dynamic state tracking.

In contrast, active retrieval policies reframe memory access as a controllable, iterative action where the model autonomously plans search trajectories and refines queries based on intermedi-

ate findings. By dynamically adjusting retrieval hyperparameters, such as the candidate count k , in response to uncertainty, active systems can emit broad queries for missing evidence or narrow searches to resolve noise. This approach enables sophisticated mechanisms like hierarchical working memory and episodic context reconstruction, allowing agents to preserve full contextual dependencies and perform selective retention that strengthens relevant traces while suppressing contradictory or stale information [7].

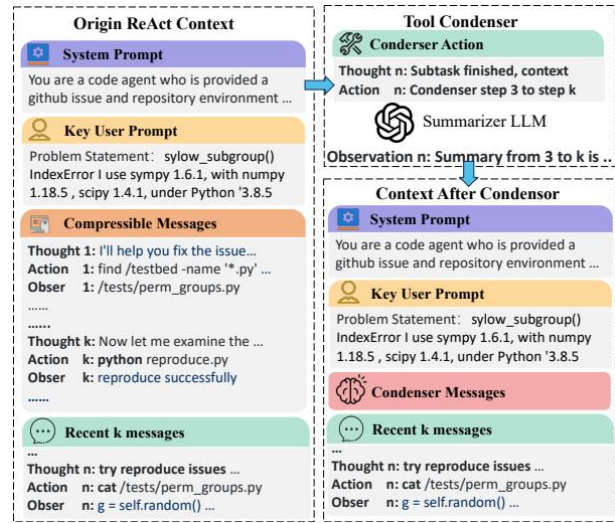


Figure 8: Chart from 'Context as a Tool: Context Management for Long-Horizon SWE-Agents' (arXiv:2512.22087)

The fundamental divergence lies in the treatment of context: passive methods prioritize lossless continuation for generation, whereas active policies optimize for lossless retrieval and agentic utility. Empirical evidence suggests that while passive expansion retains an advantage in tasks requiring precise factual recall over complex, uninterrupted histories, active retrieval excels in maintaining persona consistency and managing cross-session persistence. Consequently, optimal deployment depends on the target application's query distribution, with active policies offering superior efficiency and reasoning stability for long-term interactions where static similarity matching fails to capture evolving semantic relevance.

2.2.3 Dual-Process Systems Balancing Speed and Deliberation

Dual-process architectures in memory-augmented language models explicitly decouple rapid, heuristic retrieval from slow, deliberative reasoning to optimize the latency-fidelity trade-off. These

systems typically implement a fast mode utilizing database-centric mechanisms like standard Retrieval-Augmented Generation (RAG) for simple factoids, ensuring real-time responsiveness with fixed per-turn read costs. Conversely, a slow mode activates for complex multi-hop reasoning, employing active control loops that manage bounded working states through episodic recursive accumulation and targeted global searches. This structural differentiation allows the system to front-load computational costs during write phases while maintaining efficient, roughly constant query charges during inference, contrasting sharply with the variable, context-length-proportional costs of monolithic long-context models.

The core mechanism governing this balance involves a cognitive controller that dynamically monitors memory sufficiency to decide between immediate response generation and further information acquisition. By enforcing capacity constraints on working-state transitions and utilizing stratified scheduling, the system prevents context overflow while facilitating incremental consolidation of contradictory or temporally disjointed information. Unlike passive streaming agents, these frameworks instantiate active System-2 protocols—such as pre-thinking, retrieving, and writing—that reconcile temporal inconsistencies and merge duplicate entries through periodic sweeps. This approach ensures that complex memory operations do not linearly degrade execution time, as fine-grained nodes activate on demand and activation fan-out is capped to maintain near-linear runtime complexity relative to touched fragments.

Empirical evaluations demonstrate that such dual-process designs significantly outperform single-mode baselines in scenarios requiring temporal reasoning and multi-session coherence, often achieving substantial accuracy gains with manageable latency overheads. While simpler designs may offer marginal speed advantages in trivial tasks, they frequently fail to maintain chronological coherence or detect contradictions over long horizons. The dual-process framework addresses these limitations by balancing response quality against computational efficiency, making it particularly suitable for production environments where both factors are critical constraints. Future iterations aim to refine this into a unified adaptive framework that seamlessly switches modes based on task complexity, thereby maximizing the util-

ity of persistent storage without incurring the prohibitive costs of exhaustive context processing.

2.3 Agent-Native Memory Operations

2.3.1 Self-Reflective Memory Construction and Critique

Self-reflective memory construction transcends passive storage by integrating generative synthesis with critical evaluation during the encoding phase. Rather than merely compressing interaction histories, advanced frameworks employ a "build-and-critique" loop where candidate memories are generated as self-contained narratives encompassing emotional states, aspirations, and temporal context. These drafts are subsequently subjected to rigorous self-reflection mechanisms that assess factual consistency, relevance, and narrative coherence before final consolidation. This dual-stage process ensures that the resulting memory traces are not only semantically rich but also robust against the hallucinations and redundancies often prevalent in naive compression strategies.

The critique component functions as an internal quality control filter, leveraging meta-cognitive capabilities to identify and rectify logical inconsistencies or conflicting information within the emerging memory graph. By explicitly generating reflection tokens or critique strings alongside memory entries, systems can distinguish between supportive evidence, contradictory observations, and causal relationships. This mechanism allows for the dynamic pruning of noisy data and the reinforcement of high-fidelity episodic details, effectively transforming raw conversational turns into structured knowledge units. Consequently, the memory repository evolves from a static log into a curated database that prioritizes information essential for long-horizon reasoning and persona consistency.

Furthermore, this self-reflective approach facilitates adaptive consolidation, where the system continuously refines its internal representation based on feedback from previous retrieval and generation attempts. By treating memory construction as an iterative optimization problem, agents can align stored experiences with evolving user goals and changing environmental contexts. The integration of such reflective loops mitigates the risk of error propagation inherent in recurrent architectures, ensuring that foundational memories remain stable despite representational drift. Ulti-

mately, self-reflective construction establishes a resilient foundation for subsequent reasoning tasks, enabling agents to maintain coherent narratives over extended interaction timelines while preserving the nuance of individual experiences.

2.3.2 Operating System Inspired Virtual Context Hierarchies

Operating system-inspired virtual context hierarchies address the physical constraints of transformer context windows by simulating virtual memory management. In this paradigm, the limited context window functions as high-speed RAM, storing immediate instructions, recent dialogue turns, and active working memory, while vast external storage acts as a disk for unlimited historical data. Systems like MemGPT implement hierarchical paging mechanisms that transparently swap fragmented context chunks between these tiers, creating an illusion of infinite context for the language model without requiring architectural modifications to the underlying transformer [8].

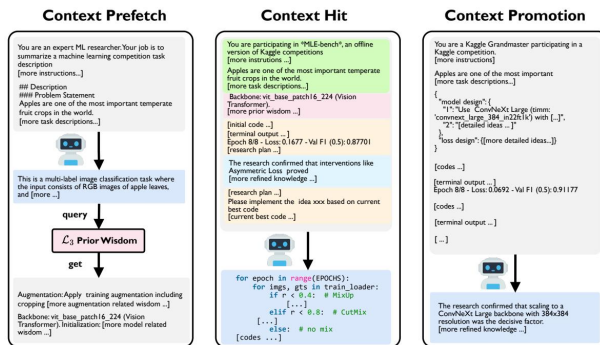


Figure 9: Chart from 'Toward Ultra-Long-Horizon Agentic Science: Cognitive Accumulation for Machine Learning Engineering' (arXiv:2601.10402)

These frameworks employ dynamic policies to manage data flow between main and external contexts, treating the context window as a scarce resource subject to strict budgeting. When the active context reaches capacity, the system identifies low-priority or dormant information based on recency and relevance, evicting it to external storage to free space for incoming tokens. Conversely, when specific historical details are required for reasoning, the system retrieves and loads relevant pages back into the main context. This approach effectively mitigates the "lost-in-the-middle" phenomenon by ensuring that only task-critical information occupies the attention mechanism's focus at any given inference step.

Despite their efficacy in extending operational horizons, current OS-inspired methods face challenges regarding temporal coherence and fragmentation. Relying on static chunking strategies often disrupts semantic continuity, as related information may be scattered across disjointed pages, complicating multi-hop reasoning tasks. Furthermore, the overhead of frequent paging operations can introduce latency, and the lack of explicit temporal modeling may lead to unstable memory states during extended interactions. Future advancements in this domain must therefore integrate more sophisticated, content-aware paging algorithms that preserve structural dependencies while maintaining the efficiency gains of hierarchical memory management.

2.3.3 Structured Fact Extraction and Entity Disambiguation

Structured fact extraction transforms unstructured dialogue into rigid schemas, such as knowledge graphs or hierarchical archives, to facilitate efficient lookup and multi-hop reasoning. This process typically employs a two-stage pipeline where large language models first identify entity types and attributes before mapping them to fixed pre-defined structures. While this approach significantly reduces storage redundancy and enhances persona continuity across sessions, it risks destructive de-contextualization by compressing complex sequential dependencies into static representations, potentially disrupting the integrity required for nuanced discourse understanding.

Entity disambiguation within these frameworks relies on resolving aliases and conflicting updates through temporal metadata and causal linking mechanisms. Advanced systems mitigate memory drift by annotating records with estimated causal parents and validity periods, allowing the model to distinguish between evolving entity states and contradictory observations. By leveraging adaptive layer weighting and damped spreading activation across semantic and temporal edges, these architectures stabilize entity-level representations, ensuring that retrieval prioritizes verifiable evidence while discarding speculative or stylistic content that could compromise factual accuracy.

However, the deployment of structured extraction involves a critical accuracy-cost trade-off specific to flat-typed fact stores. Although graph-based methods improve retrieval precision for complex queries, the overhead of index construc-

tion and maintenance often renders them less suitable for fluid, open-domain interactions compared to long-context inference. The break-even point between memory-augmented structured retrieval and raw context processing shifts based on interaction length, suggesting that rigid structural enforcement is most beneficial for tasks requiring explicit entity boundary enforcement and long-term state tracking rather than single-session information retrieval.

3 Reinforcement Learning for Memory Optimization

3.1 Reward Shaping for Long-Horizon Credit Assignment

3.1.1 Dense Auxiliary Rewards from LLM Referees

Recent advancements in reinforcing large language model reasoning have shifted from sparse, outcome-only signals to dense auxiliary rewards generated by LLM referees. Traditional reinforcement learning often suffers from credit assignment issues in long-horizon tasks due to binary terminal feedback. To mitigate this, contemporary frameworks deploy LLM-based evaluators to provide step-wise supervision, verifying intermediate artifacts such as memory tags, callback structures, and formatting compliance. This granular feedback loop ensures that agents adhere to strict syntactic requirements throughout the generation process, significantly stabilizing early-stage training dynamics.

The architecture of these dense rewards typically decomposes the objective into format adherence and semantic correctness components. Format rewards act as a gating mechanism, penalizing deviations from expected structural templates at every timestep, which prevents the propagation of unparseable states. Simultaneously, semantic rewards are derived from LLM judges that assess the quality of intermediate reasoning steps against ground truth or logical consistency criteria. By assigning scalar scores to specific actions rather than waiting for episode completion, these systems transform sparse learning signals into a continuous landscape, enabling policies to distinguish between partially correct and wholly erroneous trajectories with greater fidelity.

Integrating these auxiliary signals into algorithms like Group Relative Policy Optimization allows for more efficient gradient updates localized

to relevant tokens. The presence of dense, referee-generated rewards accelerates convergence by providing immediate correction for formatting errors and logical fallacies before they compound. Furthermore, this approach facilitates fairer credit assignment in multi-session environments, where global trajectory rewards are complemented by local, session-specific attributions. Consequently, the reliance on human preference modeling diminishes as rule-based and LLM-judged dense rewards prove superior in incentivizing complex, self-reflective reasoning capabilities without compromising the model’s native generalization.

3.1.2 Outcome-Based versus Process-Level Supervision Signals

Outcome-based supervision relies on sparse, episode-level rewards derived from final task success, which often fails to guide agents through complex, long-horizon reasoning trajectories. While effective for simple tasks, this signal suffers from high variance and credit assignment ambiguity, particularly when intermediate memory states vary significantly across rollouts. Consequently, models trained solely on global outcomes may struggle to distinguish between beneficial memory operations and coincidental successes, leading to inefficient exploration and sub-optimal policy convergence in environments requiring sustained state tracking.

To mitigate these limitations, process-level supervision introduces dense, step-wise signals that evaluate the quality of intermediate state updates and memory manipulations. By attributing rewards to specific actions such as information retrieval, memory consolidation, or constraint verification, this approach provides fine-grained guidance that shapes agent behavior toward greater efficiency. Techniques like hindsight credit assignment and information-gain supervision further refine this process by back-propagating utility to source memory items, ensuring that the model learns structured, task-aligned patterns of memory usage rather than relying on heuristic shortcuts.

Empirical analyses suggest that a hybrid strategy combining both signal types yields the most robust performance. Integrating step-level state rewards with global outcome metrics allows the agent to optimize for immediate correctness while maintaining alignment with long-term objectives. This balance prevents the model from overfitting to local heuristics that might distract from final

goal achievement, effectively addressing the trade-off between exploring diverse trajectories and adhering to valid procedural formats. Ultimately, the fusion of dense process feedback with sparse outcome validation creates a comprehensive training framework capable of supporting sophisticated, multi-turn agent interactions.

3.1.3 Counterfactual Rollouts for Memory Utility Estimation

Counterfactual rollouts address the critical challenge of attributing credit to specific memory operations within reinforcement learning frameworks. Traditional methods often suffer from biased comparisons because distinct trajectories modify intermediate memory states differently, rendering standard reward normalization ineffective. By generating local rerollouts that originate from identical intermediate memory configurations, this approach isolates the causal impact of memory retrieval and updates, ensuring fair group comparisons and cleaner supervision signals for policy optimization.

The methodology involves constructing session-wise attribution signals by comparing trajectories where only the memory interaction varies while the environmental state remains constant. This technique mitigates the issues inherent in trajectory-level rewards, where divergent memory evolutions lead to unfair credit assignment. By holding the context fixed at the point of decision, the system can accurately quantify the utility of a specific memory callback or update policy, distinguishing between genuine informational gain and stochastic environmental variance.

Empirical analyses demonstrate that leveraging counterfactual rollouts significantly enhances the stability and efficiency of memory-driven agents. This strategy allows models to learn robust policies for information retention and compression without being confounded by unrelated trajectory deviations. Consequently, agents trained with this estimation technique exhibit superior performance in long-horizon tasks, effectively balancing the trade-off between memory overhead and the precision of recalled information required for complex reasoning [9].

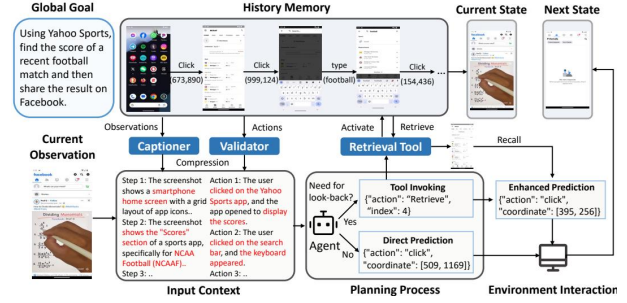


Figure 10: Chart from 'PAL-UI: PLANNING WITH ACTIVE LOOK-BACK FOR VISION-BASED GUI AGENTS' (arXiv:2510.00413)

3.2 Policy Optimization for Memory Actions

3.2.1 Group Relative Policy Optimization for Memory Operations

Group Relative Policy Optimization (GRPO) serves as the foundational algorithm for optimizing memory operations, leveraging group-wise sampling to estimate advantages without requiring a separate critic model [10]. In this framework, the policy generates a cohort of G distinct response trajectories for a given input, allowing the system to compute relative rewards within the group [11]. This mechanism is particularly effective for memory-intensive tasks where absolute reward signals are sparse, as it transforms cross-trajectory performance differences into dense, learnable gradients that guide the agent toward superior memory retention and retrieval strategies.

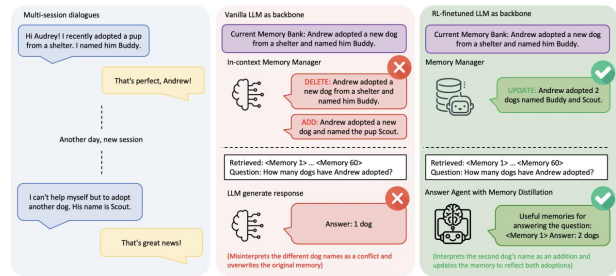


Figure 11: Chart from 'Memory-R1: Enhancing Large Language Model Agents to Manage and Utilize Memories via Reinforcement Learning' (arXiv:2508.19828)

To address the specific challenges of multi-step memory updating, where intermediate states diverge across trajectories, advanced variants employ a dual-branch optimization strategy combining global and local rollouts. The global branch maintains end-to-end optimization over full task sequences to preserve long-horizon dependencies, while the local branch utilizes resampled turns to

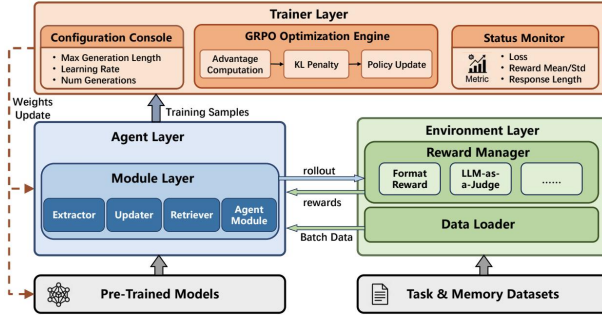


Figure 12: Chart from 'MemFactory: Unified Inference & Training Framework for Agent Memory' (arXiv:2603.29493)

reduce gradient variance associated with specific memory actions. This hierarchical approach ensures that credit assignment remains accurate even when agents encounter fragmented experiences or discontinuous rewards during complex memory synthesis and state update processes.

Furthermore, the optimization process incorporates strict attention masking mechanisms to ensure policy gradients are computed only over valid memory contexts. By restricting token attention to retained memory elements at each generation step, the system prevents leakage from evicted or irrelevant information, thereby stabilizing training dynamics. This precise control over the attention landscape, coupled with rank-based sampling of high-quality trajectories, enables the model to efficiently learn robust memory policies that balance context window constraints with the necessity of preserving critical historical information for downstream reasoning.

3.2.2 Evolutionary Strategies for Adaptive Memory Structures

Recent advancements in adaptive memory have shifted from static, heuristic-based update rules toward dynamic, learning-driven evolutionary strategies. Traditional approaches often rely on predefined structures or simple retrieval-augmented mechanisms that struggle with scalability and the efficient prioritization of information as memory volume grows. In contrast, contemporary frameworks employ reinforcement learning to optimize the entire memory lifecycle, enabling agents to autonomously determine what to store, update, or discard based on task-specific rewards and context relevance. This paradigm treats memory not merely as an external database but as an intrinsic, mutable component of the agent's reasoning pro-

cess, allowing for continuous self-evolution without explicit human intervention.

A critical innovation in this domain is the formulation of memory state transitions that incorporate both current context and historical memory states to guide evolutionary updates. Unlike conventional agents where the next memory state depends solely on immediate inputs, advanced models define states as composite tuples of existing memory and query contexts. This design facilitates sophisticated operations such as selective consolidation, where new insights are integrated with prior knowledge through probabilistic gating or atomic operations. By leveraging trajectory-level advantages and outcome-based rewards, these systems mitigate reward sparsity and attribution ambiguity, ensuring that memory modifications directly contribute to improved decision-making and long-horizon task performance.

Furthermore, evolutionary strategies now encompass the optimization of memory architecture itself, moving beyond fixed-length buffers to hierarchical and multi-structure frameworks. These systems dynamically select between complementary memory types, such as strategic, procedural, and tool-based memories, adapting their structural configuration to the complexity of the environment. Through meta-optimization and self-evolving mechanisms, agents can refine reusable memory operations over time, capturing high-value patterns that generalize across diverse tasks. This flexibility allows for robust adaptation in sparse-reward settings and complex domains, effectively bridging the gap between short-term working memory constraints and the need for persistent, scalable long-term knowledge retention.

3.2.3 Joint Optimization of Reasoning and Memory Updates

Recent advancements have shifted memory management from static retrieval mechanisms to adaptive policy optimization, where reasoning and memory updates are jointly trained via reinforcement learning. Traditional backpropagation fails to effectively govern latent memory tokens updated through discrete overwrite rules, necessitating algorithms that treat read-write-read loops as sequential decision processes. By framing memory consolidation as an integral part of the reasoning trajectory, modern frameworks enable agents to learn task-adaptive policies that dynamically determine what information to retain, discard, or syn-

thesize based on downstream task success [12].

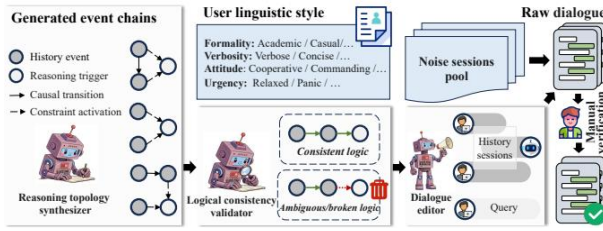


Figure 13: Chart from 'ActMem: Bridging the Gap Between Memory Retrieval and Reasoning in LLM Agents' (arXiv:2603.00026)

To address the challenge of sparse and delayed supervision inherent in long-horizon tasks, contemporary approaches employ multi-step decision processes that decompose sessions into manageable chunks for iterative refinement. Methods utilizing Group Relative Policy Optimization (GRPO) directly reward memory states that contribute to correct final answers, thereby bridging the gap between explicit answer supervision and implicit memory state evolution. This end-to-end optimization stabilizes the learning curve, ensuring that the model captures underlying patterns for memory synthesis while avoiding fragmented construction strategies that often plague systems with decoupled memory and reasoning modules.

The integration of these joint optimization strategies yields significant improvements in reasoning accuracy with negligible computational overhead. Empirical analyses indicate that while explicitly storing intermediate memories introduces marginal latency, the resulting reduction in error rates across out-of-distribution input lengths is substantial. By unifying short-term working memory operations with long-term consolidation within a shared internal state, these architectures achieve robust performance in complex, multi-turn environments. Consequently, the synergy between reasoning dynamics and memory updates represents a critical paradigm shift, transforming memory from a passive storage bank into an active, learnable component of the reasoning engine.

3.3 Experiential Learning and Skill Distillation

3.3.1 Trajectory Reuse and Canonical Demonstration Retrieval

Trajectory reuse mechanisms in long-horizon agent systems have evolved from rigid concatenation and sliding window approaches to sophis-

ticated retrieval-augmented frameworks. Early methods often lacked scalability by treating context as a static buffer, whereas modern architectures organize interaction history into episodic units or canonical demonstrations. These units encapsulate semantically related pages, including dialogue content, timestamps, and continuity links, serving as the fundamental atoms for memory organization. By parsing raw traces into structured schemas containing concise summaries and dense embeddings, systems can efficiently retrieve relevant trajectories that reflect similar goals or interaction patterns, thereby mitigating the inefficiency of re-processing redundant reasoning cycles.

To address the sparse supervision bottleneck inherent in long-context reasoning, advanced frameworks employ multi-level rewarding strategies that combine trajectory-level outcome signals with fine-grained, step-level state rewards. This approach treats memory updates not merely as context processing but as an optimizable policy component, where the information gain of each transformation is strictly evaluated. Techniques such as hindsight credit assignment attribute the utility of complex search paths back to source memory items, enabling the agent to learn reusable skills that capture type-level regularities rather than overfitting to single trajectories. Consequently, the agent can distinguish between critical search paths and explorations, effectively compressing historical interactions into executable standard operating procedures.

Furthermore, the integration of callback-retrieval mechanisms facilitates non-linear reasoning by allowing agents to query historical memories based on evolving state representations [12]. Unlike implicit compression in feature space, these methods maintain human-readable, token-level summaries that can be inspected and edited, enhancing transparency and control. By leveraging algorithms like Personalized PageRank for retrieval, systems enable pattern completion and multi-hop reasoning across temporally distributed facts without requiring parametric retraining. This synthesis of episodic retrieval and semantic consolidation ensures that high-utility experiences are abstracted into durable knowledge, allowing agents to adapt to new tasks by recalling canonical demonstrations that align with current environmental constraints and objectives.

3.3.2 Procedural Memory Consolidation from Successful Episodes

Procedural memory consolidation in large language model agents focuses on distilling successful interaction trajectories into compact, reusable operational skills rather than retaining raw episodic data. This mechanism transforms specific, high-reward sequences into generalized policies that guide future decision-making, effectively shifting the learning paradigm from rote memorization to skill acquisition. By isolating the critical actions within successful episodes, agents can encode efficient workflows that bypass redundant exploration, thereby optimizing performance in complex, long-horizon tasks where step-by-step reasoning is prone to error propagation.

The consolidation process typically employs reinforcement learning to identify and reinforce specific memory operations, such as selective updating, filtering, and retrieval, based on outcome-driven rewards. Instead of relying on dense, continuous feedback, these systems leverage sparse signals derived from task completion to strengthen the neural pathways associated with effective memory management strategies. Through multi-stage training protocols, agents learn to dynamically invoke these consolidated procedures, enabling them to maintain context coherence and execute non-linear reasoning without being overwhelmed by the computational burden of expanding context windows or unstructured history logs.

Empirical evidence suggests that this approach significantly enhances agent robustness by mitigating common failure modes like repetitive looping and information loss due to overwriting. By converting transient successes into stable procedural knowledge, agents achieve a form of continual self-improvement where each solved episode contributes to a shared skill bank applicable across diverse scenarios. This evolution from episodic recall to procedural execution represents a critical advancement in agentic architectures, allowing for scalable memory usage that preserves high-density behavioral guidance while drastically minimizing the contextual load required for inference.

3.3.3 Self-Evolving Agents via Reflection and Error Correction

Self-evolving agents leverage reflection mechanisms to transcend static policy limitations by iteratively analyzing past trajectories for error correction. Unlike traditional reinforcement learning

that optimizes solely on final rewards, these systems employ natural-language reflection to decompose failure modes within multi-turn interactions. By generating explicit critiques of their own reasoning chains, agents can identify logical inconsistencies or hallucinated facts, effectively converting negative experiences into structured learning signals without requiring parameter updates.

To mitigate accumulated errors caused by forward-only processing bottlenecks, advanced architectures integrate history-augmented states with dynamic memory retrieval. When an agent detects a deviation from grounded truth, it issues callback queries to search its episodic memory, retrieving relevant past observations to reconstruct the correct context. This capability allows the system to revisit and incorporate discarded evidence, transforming the memory bank from a passive storage unit into an active component of the reasoning loop that supports non-linear trajectory optimization.

The synergy between reflection and memory management facilitates a form of runtime reinforcement learning where agents refine their behavioral policies through experience replay. By storing corrected trajectories and feasibility feedback, these systems automatically tune their prompting strategies and decision-making heuristics for future tasks. This process enables agents to generalize from specific mistakes, avoiding repeated failures in long-horizon tasks and progressively enhancing their autonomy and robustness in complex, dynamic environments.

4 Context Compression and Management Strategies

4.1 Summarization and Abstraction Techniques

4.1.1 Goal-Oriented State Compression for Working Memory

Goal-oriented state compression transforms working memory from a passive recency buffer into an active, task-aligned representation. Unlike static summarization techniques that indiscriminately condense history, this approach selectively preserves semantic outcomes and subgoal achievements while discarding transient execution details. By mapping raw interaction trajectories into compact executive abstractions, systems maintain high-fidelity task states within fixed context windows, effectively decoupling decision-making ca-

pability from unbounded context growth.

Recent architectures implement this through dynamic, execution-driven mechanisms that optimize the trade-off between information density and fidelity. Methods such as recursive consolidation and pointer-based extraction allow agents to archive resolved reasoning steps into indexed memory slots while retaining only critical evidence in the active working set. This segmentation enables the model to focus computational resources on immediate planning requirements, utilizing compressed historical states as anchors for long-horizon coherence rather than storing verbose action-observation streams.

The integration of reinforcement learning further refines these compression policies by treating memory management as a learnable action space. Agents are trained to identify boundary-aware compression points, ensuring that salient information required for future retrieval is preserved despite the lossy nature of summarization. This symbiotic decoupling allows lightweight curator models to manage context evolution independently of the primary reasoning engine, significantly reducing token consumption while sustaining performance across complex, multi-step tasks where traditional full-context reliance fails.

4.1.2 Recursive Summarization and Gist Memory Formation

Recursive summarization has emerged as a fundamental mechanism for managing long-horizon interactions, evolving from static periodic compression to dynamic, agent-driven processes. Early approaches utilized fixed-window sliding mechanisms or heuristic pruning to fit information within context limits, often resulting in the loss of semantically critical dependencies. Contemporary frameworks address this by treating summarization as a continuous, iterative operation where completed sub-goals are abstracted into compact memory units. This recursive process allows agents to maintain a coherent narrative of task progress while strictly bounding the active context size, effectively transforming raw interaction traces into structured, high-level representations that preserve causal relevance for future decision-making.

The formation of "gist memory" represents a paradigm shift from lossy text compression to the learning of dense, task-aligned state representations. Unlike generic summarizers that merely

condense text, advanced systems optimize memory models to retain only information with deferred value, ensuring that recalled summaries directly improve downstream performance. By internalizing memory operations through reinforcement learning, agents learn to act as policies over context manipulation, executing specific actions such as peeking, partitioning, and aggregating. This enables the model to distinguish between transient noise and essential structural knowledge, creating a dependency-aware memory graph that supports complex reasoning without the computational overhead of re-processing entire historical trajectories.

Recent advancements further integrate these summarization capabilities with structured context folding and multi-scale retrieval to enhance scalability. Architectures now employ coupled operations-such as growing raw traces, compressing completed segments, and revising boundaries-to maintain state integrity while isolating flawed execution paths. This approach facilitates efficient credit assignment during training by branching memory calls rather than replaying full episodes. Consequently, the language model transitions from a passive reader of massive prompts to an active curator of its own working memory, achieving significant context compression rates while sustaining high accuracy in tasks requiring extensive temporal reasoning and multi-step planning.

4.1.3 Plan-Aware Context Pruning and Rewriting

Plan-aware context pruning and rewriting represents a paradigm shift from passive, recency-based compression to active, goal-directed memory management. Unlike traditional sliding-window heuristics or lossy summarization that indiscriminately discard historical evidence, this approach leverages explicit planning modules to identify and retain only the information critical for current subgoals. By decomposing complex tasks into hierarchical subtasks, agents can treat completed subgoals as discrete memory chunks, allowing for the aggressive pruning of intermediate reasoning traces while preserving the structural integrity of the overall plan [13].

The core mechanism involves dynamically rewriting the working context based on the execution state of the planned trajectory. When a sub-objective is finalized, the system compresses the associated token-intensive tool calls and reason-

$$v_k \Rightarrow \hat{v}_k,$$

Figure 14: Chart from 'MemoBrain: Executive Memory as an Agentic Brain for Reasoning' (arXiv:2601.08079)

ing steps into concise, pointer-heavy summaries or indexed experience entries. This process effectively decouples the growth of the context window from the length of the interaction history, transforming linear accumulation into a bounded, state-adaptive support context. Consequently, the agent maintains a compact working memory that exposes high-level plan status and essential decision-critical states without being overwhelmed by redundant past details.

Advanced implementations further refine this strategy by integrating context operations directly into the agent's policy loop, enabling fine-grained control over memory via atomic actions such as folding, branching, and selective deletion. This active curation prevents the "lost-in-the-middle" phenomenon and mitigates reasoning collapse in long-horizon tasks by ensuring that the context window remains focused on immediate dependencies and future planning requirements. Empirical results indicate that such plan-aware strategies significantly reduce token consumption while sustaining or improving task success rates, demonstrating that effective context management is a distinct cognitive skill essential for scalable autonomous agents.

4.2 Multi-Agent Collaborative Context Processing

4.2.1 Divide-and-Conquer Strategies for Distributed Reasoning

Divide-and-conquer strategies address the context bottleneck in long-horizon reasoning by decomposing complex tasks into manageable sub-problems that can be solved independently or in parallel [13]. Approaches such as Least-to-Most prompting and Plan-and-Solve explicitly structure this decomposition, guiding models to break down intricate queries into sequential sub-tasks where the solution to each step informs the next. This modularization not only simplifies the cognitive load on individual reasoning steps but also facilitates distributed execution across multiple agents

or processing units, thereby mitigating the "lost-in-the-middle" phenomenon often observed in monolithic context windows.

To maintain coherence across these fragmented sub-tasks, advanced frameworks employ structured memory mechanisms that track logical dependencies rather than mere temporal recency. Systems like ContextWeaver organize interaction traces into directed acyclic graphs, linking reasoning nodes based on information flow to support branching and merging of parallel lines of inquiry. Similarly, recursive context-aware methods utilize tree structures to inject refined subtask contexts dynamically, ensuring that relevant historical data is preserved while redundant or obsolete information is pruned. This dependency-based construction prevents context overflow and allows agents to retrieve precise evidence without triggering unnecessary tool calls.

Furthermore, these strategies integrate selective summarization and reinforcement learning to optimize the trade-off between context fidelity and computational efficiency. By decoupling reasoning from observation accumulation, agents can distill intermediate findings into compact summaries that preserve causal relationships essential for downstream decisions. This process is often governed by policies that determine what information to archive versus discard, enabling stable reasoning continuity even under strict token budgets. Ultimately, effective divide-and-conquer architectures transform unbounded reasoning challenges into scalable, graph-structured operations that sustain performance over extended interaction horizons.

4.2.2 Manager-Worker Architectures for Context Synthesis

Manager-worker architectures address context inefficiency in long-horizon tasks by decoupling high-level planning from low-level execution. In these frameworks, a central manager module synthesizes subgoals and orchestrates task decomposition, while specialized worker modules execute specific actions using localized, full-context windows. Systems like Lumos and XAgent exemplify this paradigm, where the manager generates abstract plans that ground workers in relevant environmental states, preventing the dilution of attention caused by monolithic context accumulation. This separation allows the system to maintain a compact global state while granting workers ac-

cess to detailed, task-specific information necessary for precise grounding.

Advanced implementations extend this architecture by treating subgoals not merely as instructions but as hierarchical memory chunks. HIA-GENT, for instance, leverages subgoals to structure working memory, enabling the agent to discard redundant trajectory details while preserving essential state dependencies [14]. By compressing past interactions into indexed experience memories or execution-state trees, these systems rewrite the active context into a pointer-heavy, compact representation. This approach mimics hardware caching mechanisms, where only the most salient evidence is retained in the prompt-resident view, significantly reducing token consumption without sacrificing the coherence required for complex reasoning chains.

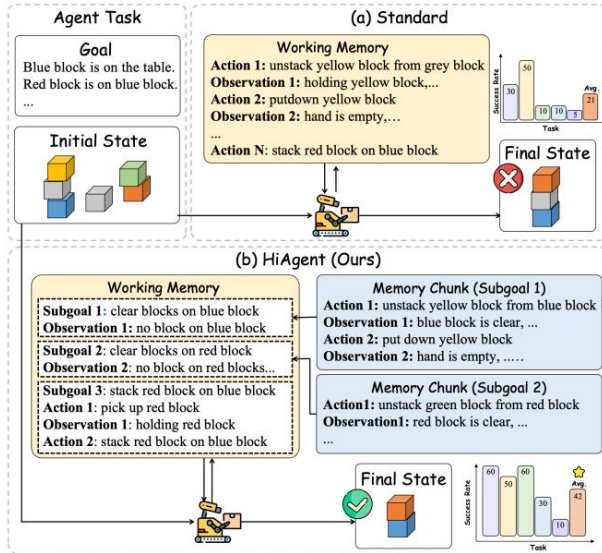


Figure 15: Chart from 'HIAAGENT: Hierarchical Working Memory Management for Solving Long-Horizon Agent Tasks with Large Language Model' (arXiv:2408.09559)

The efficacy of manager-worker synthesis lies in its ability to transform context management from a passive storage problem into an active, decision-dependent control variable. Unlike static sliding-window heuristics that indiscriminately drop older tokens, these architectures dynamically curate information density through operations such as branching, folding, and summarization. Empirical results indicate that such symbiotic frameworks substantially reduce average step counts and runtime while improving task success rates in domains requiring sustained tool

use. By isolating erroneous segments and validating compressed states through modular interaction, manager-worker designs offer a scalable solution to the context bottleneck, ensuring robust performance even under tight computational budgets.

4.2.3 Parallel Exploration and Consensus Mechanisms

Parallel exploration strategies address the limitations of linear reasoning by enabling agents to branch into multiple independent investigation paths simultaneously. Unlike sequential ReAct loops that suffer from state fragmentation and anchoring bias, modern frameworks employ breadth-first structures where related but distinct explorations originate from a shared understanding node. This architecture allows agents to evaluate diverse hypotheses in parallel, significantly reducing the risk of premature convergence on suboptimal strategies. By maintaining separate execution threads for concurrent subgoals, these systems effectively manage delayed dependencies and evolving observations, ensuring that complex, multi-hop tasks are decomposed into manageable, non-interfering streams of inquiry.

To synthesize findings from these divergent paths, consensus mechanisms act as critical coordination layers that aggregate partial results into a coherent global state. These mechanisms often utilize confidence-gated commits or voting protocols to reconcile conflicting information gathered across parallel branches before advancing the primary trajectory. By explicitly tracking cross-step strategic judgments, the system can identify redundant exploration patterns and prune unproductive sub-chains through rollback operations. This process not only mitigates the computational overhead of maintaining full context for every branch but also enhances decision stability by filtering out noise and reinforcing high-confidence insights derived from multiple independent sources.

The integration of parallelism with consensus-driven state management fundamentally transforms long-horizon agent performance by preserving execution state integrity. Rather than relying on verbose intermediate observations, these approaches distill parallel outcomes into compact thought representations that capture structural information while discarding execution-level details. This selective retention prevents context inefficiency and supports bounded explicit dereferenc-

ing, allowing the agent to scale its effective reasoning capacity without exceeding token limits. Consequently, the synergy between branching exploration and rigorous consensus validation enables robust navigation through deep dependency chains, fostering resilient behavior in dynamic environments where single-path strategies frequently fail.

4.3 Structural and Parametric Context Orchestration

4.3.1 Version-Controlled Context Management Systems

Version-controlled context management systems address the degradation of agent performance in long-horizon tasks by treating interaction history as a mutable, structured repository rather than a static token stream. Unlike traditional sliding-window approaches that indiscriminately discard older information, these systems implement explicit boundary monitoring and diagnostic logging to preserve subgoal integrity. By validating summaries before they become trusted memory, such frameworks can detect logical inconsistencies early and trigger selective corrections, effectively restoring specific trajectory segments without requiring full context re-ingestion.

These architectures often decouple context curation from task execution, employing specialized modules to actively manage information density within fixed budgets. Inspired by hardware caching and database indexing, they maintain a compact, prompt-resident "context map" that stores transferable knowledge and causal dependencies distinct from raw conversation logs. This design allows agents to dynamically compress completed sub-trajectories, flush low-utility elements, and prioritize high-salience reasoning backbones, ensuring that the working context remains logically sufficient for decision-making while minimizing computational overhead.

Furthermore, advanced implementations utilize dependency-aware construction to maintain the structural relationships between intermediate states, which is critical for complex, multi-step workflows where decisions rely on sparse, causally essential anchors. By framing context management as an active, sequential decision-making problem rather than a passive retrieval task, these systems enable state-adaptive support contexts that evolve with the task. This paradigm

shift from simple truncation to versioned, error-resilient memory structures significantly enhances tool-use reliability and coherence in environments requiring sustained, multi-session interaction.

4.3.2 Continuous Memory Tokens via Implicit Autoencoders

Continuous memory tokens via implicit autoencoders represent a paradigm shift from discrete summarization to dense, differentiable state compression. Unlike traditional methods that generate explicit textual summaries, this approach leverages learnable summary embeddings inserted at regular intervals within the token stream. These embeddings function as bottleneck layers in an implicit autoencoder architecture, forcing the model to distill high-dimensional contextual information into compact latent representations. By optimizing these vectors end-to-end, the system captures salient semantic features without the overhead of generating and re-encoding verbose natural language descriptions.

The mechanism operates by directly constructing internal memory states, such as key-value caches, from the layer-wise representations of these summary embeddings. Instead of maintaining a full-context cache that scales linearly with input length, the model attends to these condensed vectors, which serve as pointers to compressed prior states. This design significantly reduces the computational footprint during inference, enabling stable performance over extended horizons. The continuous nature of these tokens allows for gradient-based optimization, facilitating seamless integration with reinforcement learning objectives to refine memory retention policies dynamically based on task-specific rewards.

Despite their efficiency, challenges remain in ensuring the robustness of these latent representations against information loss during aggressive compression. Current implementations demonstrate that while token consumption plateaus even at extreme context lengths, the fidelity of retrieved information can degrade if the embedding capacity is insufficient for complex reasoning trajectories. Future advancements aim to balance the trade-off between compression ratios and semantic preservation, potentially incorporating self-supervised objectives like future prediction to enhance the data efficiency of training these implicit memory modules.

4.3.3 Program-Guided State Tracking and Belief Updates

Program-guided state tracking addresses the context bottleneck in long-horizon agents by replacing monolithic history logs with dynamic, structured representations. Instead of retaining all raw action-observation pairs, these systems employ hierarchical state trees or multi-scale summaries to maintain a compact yet logically sufficient working context. By treating the interaction history as a partially observable Markov decision process, agents actively prune noise and fold completed sub-trajectories into distilled state anchors, ensuring that the context window remains within token limits while preserving critical reasoning dependencies.

Belief updates in this paradigm are driven by explicit programmatic operators rather than passive accumulation. Mechanisms such as "Prune&Write" or "Grow-Compress" cycles enable fine-grained editing of the mental workspace, where the agent selectively retrieves evidence and integrates new observations into existing state nodes. This approach allows the model to reconcile dynamic context updates with stable policy learning, effectively transforming fragmented execution traces into coherent state representations. The system continuously refines its internal belief state by validating execution outcomes and annotating them with reliability signals to guide future context selection.

Furthermore, these methods facilitate active context management where the agent determines when and how to intervene in its own memory structure [15]. By leveraging trajectory segmentation and dependency summarization, the agent constructs state-adaptive support contexts that expose only the information most relevant for the current decision step. This program-guided framework not only mitigates the redundancy inherent in sliding-window baselines but also supports robust error isolation and recovery. Consequently, agents can sustain consistent reasoning over extended horizons, balancing the trade-off between information retention and computational efficiency without relying on costly re-training or external validators [16].

5 Future Directions

Despite significant advancements in architectural designs, lifecycle management, and agent-native

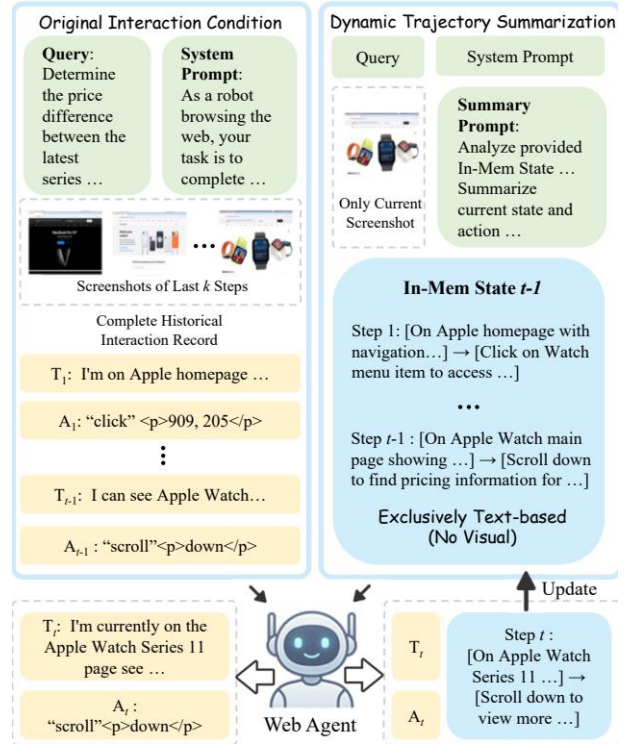
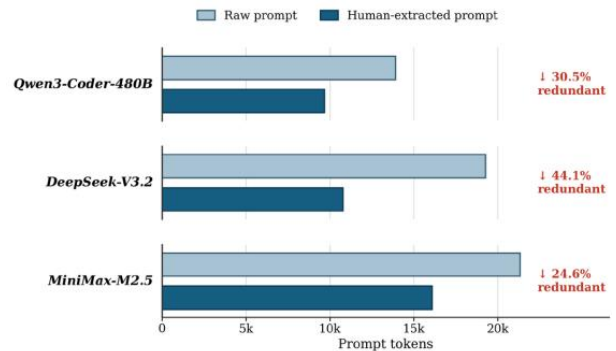


Figure 16: Chart from 'M²: Dual-Memory Augmentation for Long-Horizon Web Agents via Trajectory Summarization and Insight Retrieval' (arXiv:2603.00503)



(a) Comparison of prompt token.

Figure 17: Chart from 'A Self-Evolving Framework for Efficient Terminal Agents via Observational Context Compression' (arXiv:2604.19572)

operations for long-horizon autonomy, critical limitations persist in current methodologies. Existing systems often struggle to balance the competing demands of high-fidelity episodic retention and computational efficiency, particularly when scaling to indefinite interaction horizons. Hierarchical and graph-based structures, while effective for structured reasoning, frequently incur prohibitive overhead during dynamic updates and suffer from consistency drift across abstraction layers. Similarly, biologically inspired forgetting mechanisms lack robust, task-agnostic criteria for determining what information to consolidate or discard, leading to either premature loss of rare but vital facts or the accumulation of redundant noise. Furthermore, the integration of self-reflection into memory construction remains computationally expensive, often creating latency bottlenecks that hinder real-time responsiveness in complex, multi-step environments.

Future research must prioritize the development of adaptive, hybrid memory frameworks that dynamically reconfigure their structural topology based on task complexity and resource constraints. Instead of relying on static architectural choices, next-generation agents should employ meta-learning strategies to optimize the trade-off between hierarchical indexing depth and retrieval speed in real time. A promising direction involves the creation of unified memory substrates that seamlessly blend symbolic knowledge graphs with neural vector stores, enabling agents to perform rigorous causal inference while maintaining the flexibility of semantic similarity search. Additionally, there is a pressing need to refine active retrieval policies by incorporating explicit uncertainty quantification, allowing agents to autonomously decide when to engage in deep, multi-hop reasoning versus relying on heuristic recall. Investigating neuro-symbolic approaches that mimic the interplay between rapid intuition and slow deliberation could further enhance the efficiency of memory governance, ensuring that computational resources are allocated only when necessary for maintaining coherence and factual accuracy.

The realization of these proposed advancements holds the potential to fundamentally transform the capabilities of autonomous agents, enabling them to operate reliably over months or years of continuous interaction. By overcoming current bottlenecks in scalability and consistency, future mem-

ory systems will empower agents to sustain complex personas, execute intricate long-term plans, and adapt to evolving user needs without succumbing to semantic decay or hallucination. This evolution from passive context buffering to active, intelligent memory governance will not only enhance performance in specialized domains such as scientific discovery and personalized assistance but also lay the foundational infrastructure for truly general-purpose artificial intelligence capable of lifelong learning and reasoning.

6 Conclusion

This survey has systematically examined the architectural, lifecycle, and operational innovations essential for enabling Large Language Models to function as effective long-horizon autonomous agents. We identified that overcoming the inherent limitations of static context windows requires a fundamental shift from passive token buffering to active memory governance. Architecturally, the transition from flat vector stores to hierarchical temporal trees and causal knowledge graphs allows agents to maintain semantic coherence and perform multi-hop inference across extended interaction histories. Furthermore, the adoption of biologically inspired lifecycle management, characterized by adaptive forgetting and hippocampal-neocortical consolidation strategies, ensures that memory systems remain scalable and resistant to interference. Finally, the integration of agent-native operations, such as self-reflective build-and-critique loops, transforms raw conversational data into curated, high-fidelity knowledge units, thereby mitigating hallucinations and reinforcing logical consistency throughout the agent's lifespan.

The significance of this work lies in its provision of a unified taxonomy that synthesizes disparate research threads into a cohesive framework for evaluating agentic memory. By distinguishing between structural complexity, dynamic consolidation mechanisms, and operational meta-cognition, this survey clarifies the specific trade-offs regarding scalability, retrieval precision, and computational overhead inherent in current methodologies. Unlike previous reviews that broadly address context extension or general retrieval augmentation, this analysis zeroes in on the unique requirements of persistent autonomy, offering critical insights into how modern systems balance information fi-

delity with efficiency. This structured perspective not only delineates the current state-of-the-art but also exposes pivotal gaps in existing approaches, serving as an indispensable reference for understanding the evolution from simple context retention to sophisticated, human-like memory retention and recall.

Looking forward, the development of truly autonomous agents demands a concerted effort to bridge the remaining gaps between theoretical memory architectures and robust, real-world deployment. Future research must prioritize the optimization of hierarchical indexing to reduce latency without sacrificing reasoning depth, while also exploring more nuanced models of forgetting that align closely with task-specific relevance rather than generic decay curves. Additionally, there is an urgent need for standardized benchmarks capable of evaluating memory performance over indefinite horizons, moving beyond short-term task completion to assess long-term persona consistency and causal reasoning capabilities. We call upon the research community to embrace these challenges by fostering interdisciplinary collaboration between cognitive science and artificial intelligence, ultimately driving the creation of next-generation agents that can sustain reliable, coherent, and efficient performance across the boundless complexities of open-ended environments.

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